Sleep Apnea

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Adapted from: Sleep Apnea

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of patients with sleep apnea.
2. Explain the pathophysiology of sleep apnea.
3. Interpret clinical findings to properly diagnose sleep apnea.
4. Outline the management plan of sleep apnea.

CASE CONNECTION

RE is a 53-year-old male who comes to the clinic accompanied by his wife. "His snoring keeps getting worse. Sometimes, it sounds as if he has stopped snoring, and I worry that he's not breathing, because he's usually so loud," she says. RE adds, "I'm not sleeping well at night, and then I keep falling asleep when I sit down to watch TV. I don't get it." On exam, RE's body mass index is 31 kg/m^2, and his blood pressure is 150/90 mm Hg. His physical exam is otherwise unremarkable.

How will you explain RE's condition to him and his wife? Consider your answer as you read, and we'll revisit RE at the end of the brick.

GO TO CONCLUSION

What Is Sleep Apnea?

Sleep apnea is a sleep disorder characterized by decreased or impaired ventilation during sleep, leading to a disruption of sleep. It's the most common sleep-related breathing disorder.

The two main types of sleep apnea are:

- Obstructive sleep apnea (OSA), more common
- Central sleep apnea (CSA), much less common

What's the difference? OSA is a chronic condition of recurrent **upper airway obstruction during sleep**. Obesity and various pharyngeal abnormalities (like enlarged adenoids in the throat) are risk factors.

In contrast, CSA is a condition in which **ventilation is decreased because of an impaired central nervous system (CNS) respiratory drive**. The symptoms of CSA are very similar to those of OSA, but the cause is quite different. It's a neurologic problem seen mostly in older patients.

Approximately 22 million people in the United States have OSA, with up to 80% of cases undiagnosed. The estimated incidence in US males is 15% to

30% and 10% to 15% in premenopausal females. After menopause, the incidence in females approaches that of males.

CLINICAL CORRELATION

OSA occurs in 1 to 5 percent of children. It can occur at any age and may be most common in those between two and six years of age.

CSA is much less common, seen in about 1% of the population.

OSA is a chronic condition of recurrent upper airway obstruction during sleep, whereas CSA is a condition in which ventilation is decreased because of an impaired respiratory drive.

How Do Patients With Sleep Apnea Present (LO #1)?

Most people with sleep apnea are unaware that they have this condition. They may have nonspecific symptoms of disordered sleep, like **excessive daytime sleepiness**, morning headaches, or fatigue. Patients with OSA may wake up gasping or choking or with a dry mouth.

The history may come from a partner they sleep with. In both OSA and CSA, the sleep partner may have noted that the patient stops breathing while asleep (apneic episode). In OSA, the partner may report heavy snoring, choking, or snorting from the patient while asleep.

Common symptoms of OSA are daytime fatigue, morning headaches, and awaking gasping or choking.

CLINICAL CORRELATION

In children, the most common symptom is habitual or loud snoring. Patients may have additional symptoms such as mouth breathing, noisy breathing, pauses in breathing, coughing or choking in sleep, restless sleep, and nighttime sweating.

What Is the Pathophysiology of Sleep Apnea (LO #2)?

In both OSA and CSA, the patient experiences hypoventilation during sleep but for different reasons.

Causes of Obstructive Sleep Apnea

In OSA, something interferes with typical air exchange (hence, “obstructive”) while the patient is asleep. This obstruction can be due to an upper airway abnormality or obesity, which narrows the upper respiratory passage.

In Figure 1, left, we see how air needs to travel for gas exchange to reach the lungs.

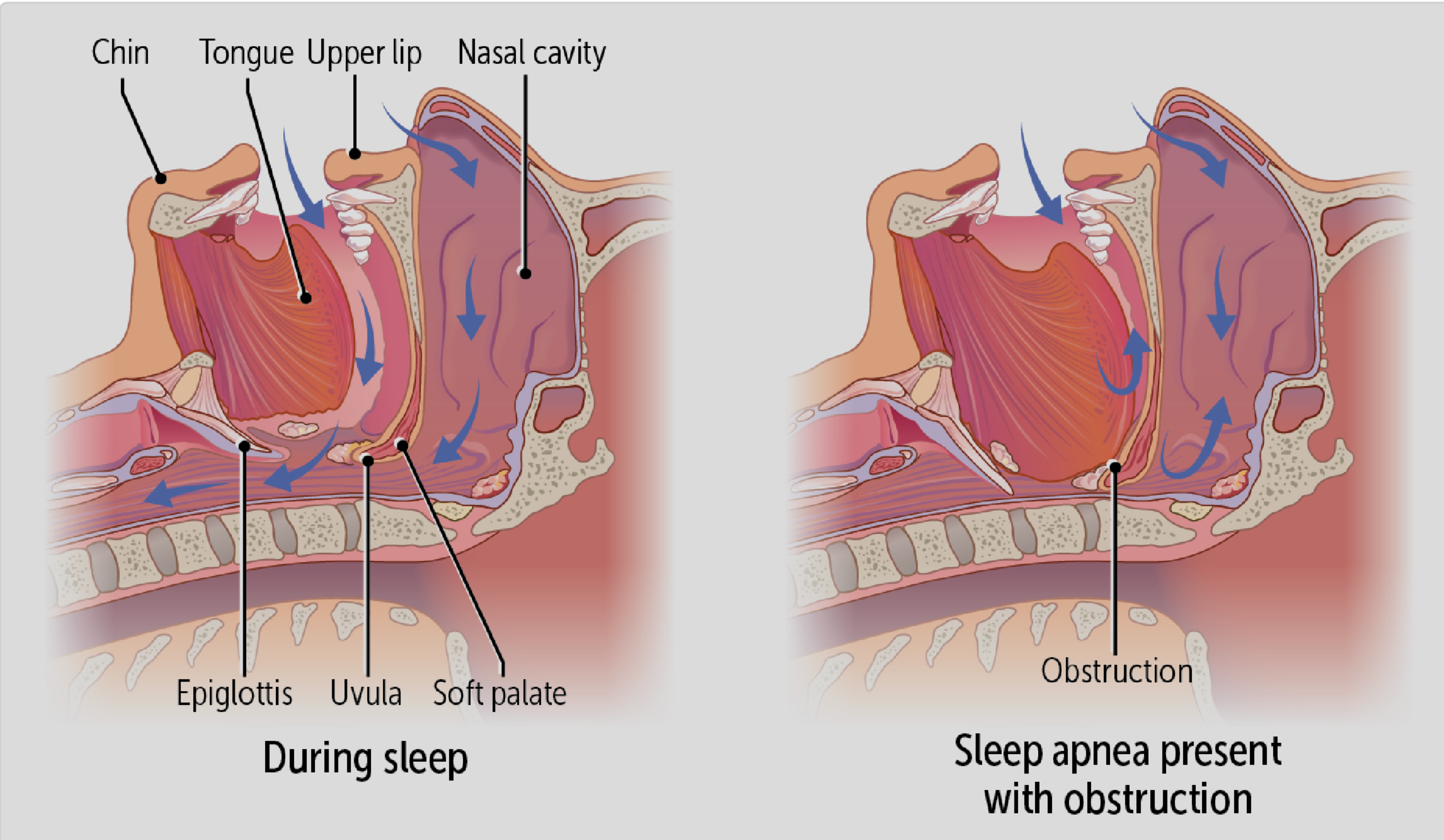

Figure 1

When a person is lying supine, air must pass over and then beneath the tongue, beneath the soft palate, and get by the epiglottis to reach the trachea. All these structures become potential obstructions to airflow in patients with OSA (Figure 1, right). This leads to increased effort to breathe but with decreased air passage through the upper airway.

Specific disorders associated with OSA include:

- **Obesity** (body mass index [BMI] >30 kg/m^2) is a common clinical finding in patients with OSA. Patients with obesity commonly have increased fat deposits around the neck that are difficult to move during inspiration. This leads to a complete or partial closing of the airway. Decreased chest wall excursion can occur because of excess fat deposition there. Note that increased waist and neck diameter are more connected to OSA than obesity itself.

- **Abnormalities of the upper airways** can also obstruct normal breathing when the patient is lying down. These may include nasal congestion, an abnormally positioned or short mandible, a large tongue or epiglottis, enlarged tonsils or adenoids, redundant neck tissue (as seen in obesity), craniofacial structural abnormalities that may limit airflow, and weakened upper airway muscles. The most common cause in children (who do not usually get OSA) is adenotonsillar hypertrophy, which is surgically correctable.
- **Other associations** include pregnancy, cardiovascular disease (hypertension, heart failure, atrial fibrillation), chronic lung disease (chronic obstructive pulmonary disease, asthma, pulmonary hypertension), endocrine disease (hypothyroidism, acromegaly, polycystic ovaries), Parkinson disease, and chronic kidney disease. For most, the reason for the association is not known.

CLINICAL CORRELATION

There is also an association between OSA and the medication gabapentin (a medication primarily used for epilepsy and nerve pain), because gabapentin is thought to decrease muscle tone to the pharyngeal muscles, leading to upper airway collapse and thus OSA.

Causes of Central Sleep Apnea

Patients with CSA have decreased respiratory drive from the brain. Increased age, male sex, heart failure, stroke, and use of opioids are risk factors.

What causes decreased respiratory drive? Let's start with the normal respiration mechanism. The brain's respiratory center is very sensitive to the partial pressure of arterial carbon dioxide ($PaCO_2$). When $PaCO_2$ decreases below a set apneic threshold, impulses will not be sent to the muscles to initiate respiration, and the patient will temporarily stop breathing. Consequently, CO_2 will accumulate in the blood, and the $PaCO_2$ will rise above the required threshold for breathing to restart.

In the most common cause of CSA, one of several triggers causes **transient hyperventilation**, lowering the $PaCO_2$ and inducing the central apnea. These patients appear to be much more sensitive to this lowered $PaCO_2$ than healthy patients, so they develop more apneic episodes. Stroke and heart failure appear to cause CSA by this mechanism, although the mechanism is not always known.

A smaller number of patients instead have persistent hypoventilation during sleep, not driven by low $PaCO_2$. This is mostly seen in patients with neuromuscular disease, like encephalitis or polio, as well as in patients taking respiratory suppressant drugs such as opiates or benzodiazepines.

CSA is a decrease in the respiratory drive, whereas OSA shows an upper airway obstruction to ventilation, increasing the work of inspiration and decreasing airflow.

Complications

Patients with chronic OSA are at increased risk for **cardiovascular diseases** such as hypertension, pulmonary hypertension, coronary artery disease, and right-sided heart failure. The precise mechanisms are not known, although intermittent hypertension during sleep as well as an activated inflammatory response in patients with OSA seem to be involved. The obesity associated with most patients with OSA may also contribute.

How Do We Diagnose Sleep Apnea (LO #3)?

The first step in diagnosis is obtaining a good patient history. Rule out other causes for disordered sleep, like a noisy household, irregular work shifts, anxiety, and psychiatric disease.

Physical Examination

No specific findings on the physical exam are diagnostic for sleep apnea. In OSA, the patient may have clinical findings of obesity (eg, BMI of >30 kg/m^2). However, OSA can also occur in patients with lower body weight. An **enlarged neck or waist circumference** is more predictive than total body weight. Some patients with OSA may have a small or posteriorly oriented mandible or narrowing of the oropharynx (eg, by enlarged adenoids, tonsils, or uvula or nasal polyps). In addition, elevated blood pressure may be present —OSA is a cause of secondary hypertension.

Diagnostic Testing

A diagnosis of sleep apnea can be made with **polysomnography** (**PSG**), a sleep study done in a sleep laboratory or at home.

To perform PSG, a patient is connected to a variety of monitors that measure brain activity, oxygen saturation, and respiratory effort as well as video the sleeping patient. The severity of OSA is based on the number of respiratory-related disturbances in an hour of sleep.

The diagnostic criteria for obstructive sleep apnea depends on whether the data was derived from a sleep laboratory or at home.

For polysomnography (PSG) performed in the sleep laboratory, OSA is confirmed if either of the two criteria below is present and the symptoms are not better explained by another current sleep disorder, medical condition, medication or substance use:

- There are 5 or more predominantly obstructive respiratory events (apneas, hypopneas, or respiratory effort-related arousals (RERAs) per hour of sleep in a patient with one or more of the following:
  - Sleepiness, fatigue, insomnia, or other symptoms leading to impaired sleep-related quality of life
  - Waking up with breath holding, gasping, or choking
  - Habitual snoring or breathing interruptions noted by a bed partner or other observer
- There are 15 or more predominantly obstructive respiratory events per hour of sleep

For home sleep apnea testing (HSAT), most devices do not include electroencephalogram (EEG) monitoring; therefore, RERAs and hypopneas characterized by arousals cannot be reliably identified. With these caveats in mind, the criteria for diagnosis is the same as PSG testing.

Severity depends on the number of respiratory disturbances during an hour of sleep, as recorded by polysomnography.

For children to be definitively diagnosed with OSA, both clinical and polysomnographic criteria should be present:

Clinical Criteria:

- The presence of one or more of the following symptoms:
  - Snoring
  - Labored, paradoxical, or obstructed breathing during sleep
  - Sleepiness, hyperactivity, behavior problems, or learning and other cognitive problems

PSG Criteria:

- The PSG demonstrates one or both of the following:
  - One or more obstructive apneas, mixed apneas, or hypopneas, per hour of sleep
  - A pattern of obstructive hypoventilation, defined as at least 25 percent of total sleep time with hypercapnia (partial pressure of carbon dioxide [$PaCO_2$] >50 mmHg) in association with one or more of the following: snoring, flattening of the nasal pressure waveform, or paradoxical thoracoabdominal motion

How Do We Manage Sleep Apnea (LO #4)?

A variety of methods to treat sleep apnea are available. Because sleep apnea is a chronic disease, the patient and provider should expect making a lifelong effort to manage the condition.

Management of Obstructive Sleep Apnea

In OSA, encouraging weight loss and avoidance of alcohol and sedatives as well as counseling about sleep positioning may be helpful. Medications to avoid include benzodiazepines, barbiturates, opiates, and antihistamines. Some patients with OSA develop more symptoms when sleeping supine, and, if documented, this can be treated with home devices that vibrate when the patient turns to that position.

Continuous Positive Airway Pressure. If lifestyle changes are unsuccessful, the best treatment for most patients with OSA is **continuous positive airway pressure (CPAP)**. Air is delivered via a mask at a pressure high enough to overcome obstruction in the head and neck and enter the trachea for air exchange. CPAP works by increasing upper airway pressure, helping to continuously maintain the patency of the upper airway (Figure 2).

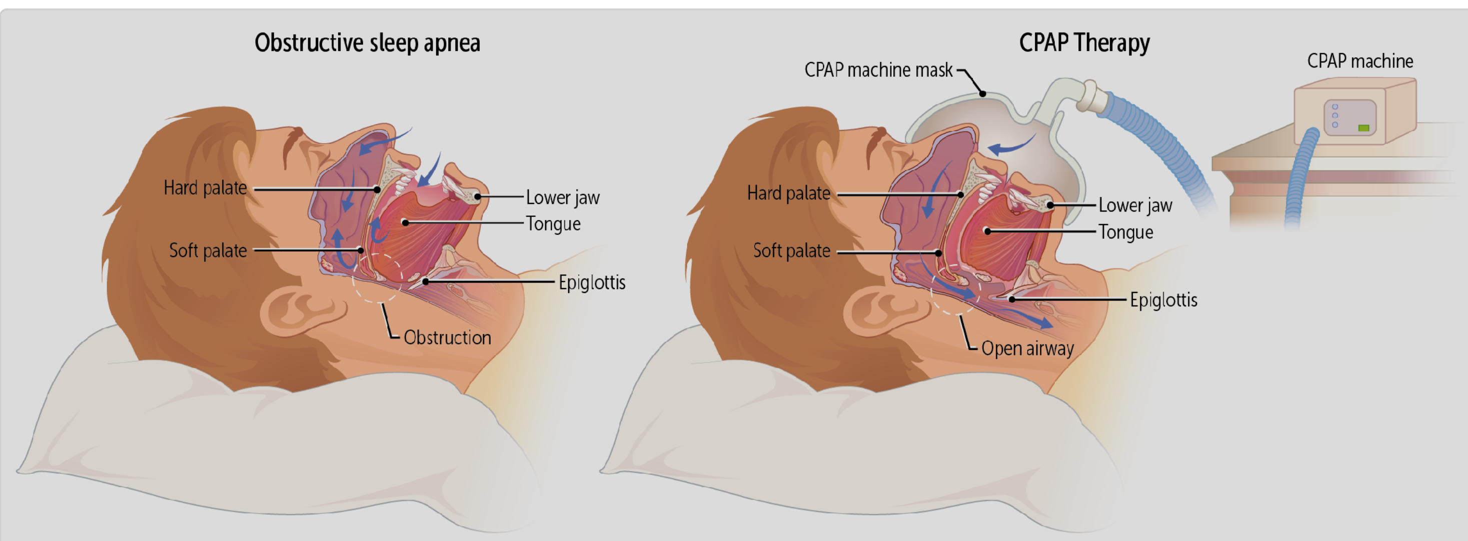

Figure 2

CPAP increases airway pressure, preventing the collapse of upper airway structures that can impede airflow, causing OSA.

Oral Appliances. In mild to moderate cases of sleep apnea, oral appliances can help keep the airways open during sleep. Depending on the patient, dental or mandibular devices can be used to keep the tongue from blocking the throat or to advance the mandible forward, thus enlarging the available space.

Surgery. Children with suspected OSA should be referred to an otolaryngologist. Adenotonsillectomy is the first-line therapy for children who have adenotonsillar hypertrophy. Even if hypertrophy of the tissue is not the singular cause of OSA, surgery has been shown to be beneficial: it improves daytime behavior, decreases sleepiness, and increases quality of life.

Although surgical correction in adults is rare, for adult patients with tonsillar or adenoid hypertrophy or structural abnormalities, surgery has been shown to be effective.

Management of Central Sleep Apnea

Treating CSA begins with removing sedative drugs, if possible. Patients with CSA due to heart failure may benefit from CPAP, like those with OSA. Patients with persistent nighttime hypoventilation (eg, after strokes) may instead benefit from bilevel positive airway pressure (BiPAP), which is like CPAP but with higher pressures delivered during inspiration.

CASE CONNECTION

BACK TO INTRODUCTION

Thinking back to RE, how do you explain his condition?

Sleep apnea is high on your differential diagnosis list, and you schedule a sleep study to confirm your suspicion. You explain to RE that this is not just a problem of disrupted sleep: it can affect his overall health, including his heart and blood pressure. You tell RE that you'll arrange a sleep study and order some tests to check on his blood pressure and heart. You also prescribe medication for his hypertension. You schedule a follow-up appointment for shortly after the sleep study when you will be able to go over the test results, get him started with sleep apnea treatment, and begin the conversation about his weight and overall health.

Summary

- Sleep apnea is decreased or impaired breathing during sleep.
- This can be due to upper airway obstruction (obstructive sleep apnea [OSA]) or less commonly due to decreased respiration signals from the brain (central sleep apnea [CSA]).

- Symptoms may include being extremely tired during the day; not feeling rested after sleep; and snoring, snorting, or choking during sleep.
- The sleep partner may have noted that the patient stops breathing while asleep.
- OSA occurs when an upper airway obstruction prevents ventilation.
- Causes of OSA include external compression of the airway by obesity or in conditions causing upper airway obstruction, such as jaw or adenoid abnormalities.
- Patients with CSA have decreased respiratory drive due to aberrant central nervous system regulation of respiration.
- Causes of CSA include stroke, heart failure, and medications.
- The first step in diagnosis is a patient history to rule out other causes for sleep-related symptoms.
- Patients suspected of having sleep apnea should undergo polysomnography (PSG) for evidence of apneic episodes.
- Patients should avoid alcohol and sedatives before sleep.
- First-line therapy for adults with OSA is use of continuous positive airway pressure (CPAP).
- First-line therapy for children with OSA is often adenotonsillectomy.

Review Questions

1. **Which of the following is not a known risk factor for central sleep apnea?**

A. Barbiturates

B. Congestive heart failure

C. Obesity

D. Opioids

E. Stroke

Explanation (requires correct answer)

2. **A 62-year-old male comes to the clinic for evaluation of fatigue. He reports being increasingly tired over the last 3 months. His wife has observed him take "long pauses" between breaths when he sleeps. He does not smoke or drink alcohol. His body mass index is 35 kg/m^2. Which of the following is the most appropriate next step in evaluation of this patient?**

A. Continuous positive airway pressure

B. Increased daily caffeine intake

C. Polysomnography

D. Selective serotonin reuptake inhibitor therapy for depression

E. Thyroid laboratory studies

Explanation (requires correct answer)

3. **Which of the following is the most significant risk factor for the development of obstructive sleep apnea?**

A. Asthma

B. Diabetes mellitus

C. Hypothyroidism

D. Increased waist circumference

E. Tobacco smoking

Explanation **(requires correct answer)**

4. Which of the following best describes how a continuous positive airway pressure mask improves symptoms in obstructive sleep apnea?

A. Decreases intrathoracic pressure to push air into the lungs

B. Increases intrathoracic pressure to pull air into the lungs

C. Increases the force with which the diaphragm contracts

D. Provides consistent negative pressure to overcome obstruction

E. Provides consistent positive pressure to overcome obstruction

Explanation (requires correct answer)

How would you rate this Brick?

References

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 **Differentiate the clinical presentation of patients with sleep apnea.**

02 **Explain the pathophysiology of sleep apnea.**

03 **Interpret clinical findings to properly diagnose sleep apnea.**

04 **Outline the management plan of sleep apnea.**

Objective Confidence: 0/4